Smart Home AIoT – Test Data Collection Sheet – Phase E – Adaptive Lighting

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Instructions for use:
1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:
- Pass Rate = (Pass Count / 10) × 100%
- Avg RT = (response times) / 10
- Variance = (xi -)² / n
- Std Dev = √variance

Coverage Matrix

Phase	Description	Count
Phase E	Phase E – Adaptive Lighting	8
	TOTAL	8

Phase E — Adaptive Lighting

E01	Bedroom Adaptive (PIR + LDR)	Phase E

Method: Setup LDR > 2200 + walk PIR

Expected: switch.bedroom_light ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E02	Living Room Adaptive (Zigbee PIR + LDR + brightness scale)	Phase E

Method: LDR > 2200 + motion

Expected: Tuya bulb ON brightness scaled by LDR (30-100%)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E03	Bathroom Adaptive (LD2410 + LDR)	Phase E

Method: LDR > 2000 + presence

Expected: switch.bathroom_light ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E04	Garage Adaptive (PIR + LDR)	Phase E

Method: LDR > 2400 + motion

Expected: garage_light + pathway_light ON

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E05 Adaptive NO-trigger (สว่างพอ) Phase E

Method: LDR < 2200 + motion

Expected: light ยัง OFF (no trigger)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E06 Bedroom Auto-off (5 min) Phase E

Method: turn on light + no motion 5 min

Expected: auto OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E07	Garage Auto-off (5 min)	Phase E

Method: turn on lights + no motion 5 min

Expected: both garage lights OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments:

E08	Bathroom Auto-off (5 min)	Phase E

Method: light on + no presence 5 min

Expected: bath light OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10	Pass Rate: ____%	Avg RT: ____ ms	Std Dev: ____ ms

Comments: